

TECHNICAL SPECIFICATION

FOR

**3 PHASE 4 WIRE CT & PT OPERATED 0.5S ACCURACY CLASS CAT "B"
STATICTRIVECTOR ENERGY METER FITTED WITH PILFER PROOF METER BOX SUITABLE
TO HOUSE THE METER & TEST TERMINAL BLOCK**

FOR

BOUNDARY METERING IN 11 KV FEEDER

1.0 SCOPE

This specification shall cover design, engineering, manufacture, assembly, inspection, testing at manufacturers works before dispatch, supply and delivery at destination anywhere in the state of West Bengal of static 3 phase 4 wire CT&PT operated 0.5S Accuracy Class Cat "B" DLMS Compliant 4 Quadrant Tri-vector Energy Meter fitted with Pilfer Proof Meter Box for Boundary Metering in 11 KV Feeder for measurement of different electrical parameters listed elsewhere in the document including Active Energy(Wh), Reactive Energy (varh), Apparent Energy(VAh) etc. over a Power Factor range of Zero Lag to Unity to Zero Lead both in Export & Import mode. The Meters shall have facility of recording tamper information & load survey of both import & export phase wise active energy, apparent energy, phase currents & phase voltages & other parameters in non volatile memory. These meters shall have communication port to interface for remote meter reading.

2.0 CLIMATIC CONDITION

The meter shall be suitably designed for satisfactory operation under hot and hazardous tropical climatic conditions and shall be dust & vermin proof. All the parts and surfaces, which are subject to corrosion, shall either be made of such material or shall be provided with such protective finish, that those shall remain protected from any injurious effect of excessive humidity.

The meter shall be suitable for satisfactory continuous operation under the following tropical conditions:-

- 2.1 Maximum ambient air temperature : 55°C
- 2.2 Minimum ambient air temperature : -10°C
- 2.3 Maximum Relative Humidity : 95% (non-condensing)
- 2.4 Minimum Relative Humidity : 10%
- 2.5 Maximum altitude above mean sea level : 3000 meters
- 2.6 Average number of tropical monsoon per annum : 5 months
- 2.7 Maximum Annual Rainfall : 1500 mm
- 2.8 Maximum Wind Pressure : 150 Kg/m. sq.

3.0 APPLICABLE STANDARDS

Unless specified otherwise elsewhere in this specification, the performance & testing of the CT&PT operated meters of accuracy class 0.5S shall conform to the following Indian / International Standards, to be read with up-to-date and latest amendments / revisions thereof as on 90 days prior to floating of tender.

Standard	Title
IS14697: 1999 read with latest amendments	Specification for AC Static Transformer operated Watt Hour & VAR Hour meters, Class 0.5S
CBIP Report No. 325 & its latest amendments, if any	Specification for AC Static Electrical Energy Meters
IS 15959: 2011 read with latest amendments	Data Exchange for Electricity Meter - Reading Tariff and Load Control - Companion Specification
IEC 62056(Part 21, 31, 61)	Electricity metering: Data exchange for meter reading, tariff and load control: Direct local data exchange, Local Area Network data exchange, Object identification system (OBIS)
IEC 62052-11	Electricity metering equipment (AC) –General requirements, tests and test conditions -Part 11 : Metering equipment
IEC 62053-22	Electricity metering equipment (AC) –Particular requirements - Part-22 : Static Meters for Active Energy (Class 0.5S)
IS-15707	Specification for Testing, evaluation, installation & maintenance of AC Electricity Meters-Code of Practice
IS 13410 :1992 with latest amendment	Specification for Pilfer Proof Meter box suitable for 3 Phase Static Energy Meter
IS 12346 :1988	Specification for testing equipment for AC Static Electrical Energy Meter (latest amendment)

4.0 GENERAL TECHNICAL REQUIREMENTS

4.1	Type	AMR Compatible Static 3 Phase4 Wire(11KV) Tri-Vector Energy Meter
4.2	Rated Frequency	50 Hz $\pm$ 5%
4.3	Accuracy Class	0.5s (Meter shall meet the same class of accuracy for reactive energy also)
4.4	Rated Secondary Voltage	63.5V (Phase to Neutral)as per PT for 11 KV feeders
4.5	Allowable Voltage Variation	70% to 120% of Reference Voltage
4.6	Rated Secondary Current (I _b)	1A as per CT for 11KV feeders
4.7	Starting Current	The meter shall start registering energy at 0.1% of basic current at unity power factor and shall be fully functional within five seconds after the rated voltage is applied.
4.8	Running at No Load	When 70% & 120% of reference voltage is applied at meter terminal and no current flows in the current circuit, the test output of the meter shall not produce more than one pulse.
4.9	Maximum Continuous Current	2 times of I _b . Starting and Short time current shall be as per IS-14697.
4.10	Power Consumption	The active and apparent power consumption in each voltage circuit at reference voltage, reference temperature and reference frequency shall be as per IS 14697, i.e, 1.5 W, 8 VA The apparent power taken by each current circuit at basic current, reference frequency and reference temperature shall be as per IS 14697, i.e, 1 VA.
4.11	Power Factor	0.0 Lag -Unity- 0.0 Lead
4.12	Design	Meter shall be designed with application specific integrated circuit (ASIC) or micro controller; shall have no moving part; electronic components shall be assembled on printed circuit board using surface mounting technology; factory calibration using high accuracy (0.05 class) software based test bench. Assembly of electronic components shall be as per ANSI /IPC-A-610 standard.

5.0 CONSTRUCTION

5.1 The case, winding, voltage circuit, sealing arrangements, registers, terminal block, terminal cover & name plate etc. shall be in accordance with the relevant standards. The meter shall be compact & reliable in design, easy to transport & immune to vibration & shock involved in the transportation & handling. The construction of the meter shall ensure consistence in performance under all conditions, especially during heavy rains / very hot weathers. The insulating materials used in the meter shall be non-hygroscopic, non-ageing & have tested quality. The meter shall be sealed in such a way that the internal parts of the meter become inaccessible and attempts to open the meter shall result in viable damage to the meter cover i.e. break-to-open type. This is to be achieved by welding all the four sides of the meter base and cover ultrasonically or by utilizing any other technology, equally or more efficacious.

5.2 The meter shall comply with latest technology such as Microcircuit or Application Specific Integrated Circuit (ASIC) to ensure reliable performance. The mounting of the components on the PCB shall compulsorily be of Surface Mounted Technology (SMT) type. Power supply component shall be of PTH type. The electronic components used in the meter shall be of high quality and there shall be no drift in the accuracy of the meter for at least ten years. The circuitry of the meter shall be compatible with 16 Bit (or better) ASIC with compatible processor and meter shall be based on digital measuring and sampling technique.

5.3 The meter shall be housed in a safe, high grade, unbreakable, fire resistant, UV stabilized, virgin Polycarbonate casing of projection mounting type. The meter cover shall be transparent / translucent. But the viewing portion shall be transparent for easy reading of displayed parameters and observation of operation indicators. The meter base may not be transparent but it shall not be black in colour. The meter casing shall not change in shape, colour, size and dimension when subjected to 72 hrs on UV test as per ASTM D 53. It shall withstand 650°C glow wire test and heat deflection test as per ISO 75 or as per IEC 60068 -2-5.

5.4 In addition to the above, the meter cover shall be sealable to the meter base with at least 2 no. of bar-coded seals bearing the identification marks of the manufacturer. Suitable arrangement shall be made for fitting/fixing of utility seal at two sides of meter terminal cover in such a manner that access to the terminal cannot be made without removing the seal. There shall be provision for sealing the optical port also.

5.5 The polycarbonate material of only the following manufacturers shall be used:

- | | | |
|-------|-----------------------|---|
| 5.5.1 | G.E. PLASIC : | LEXAN 943A or equivalent like 943, 123R for meter cover & terminal cover
LEXAN 503R or equivalent like 500, 143R for meter base and terminal block |
| 5.5.2 | BAYER : | Grade corresponding to above |
| 5.5.3 | DOW Chemical : | --do-- |
| 5.5.4 | MITSUBISHI : | --do-- |
| 5.5.5 | TEJIN : | --do-- |
| 5.5.6 | DUPONT : | --do-- |

6.0 METER CASE AND COVER

6.1 In case, ultrasonic welding is done using plate / strip, the material of plate / strip shall have to be same as that of cover and base. The manufacturer's logo shall be embossed on the plate / strip. The material of the meter body (case and cover) shall be made of Engineering Plastic.

6.2 The meter cover shall be fixed to the meter base (case) with unidirectional screws, so that the same cannot be opened by use of screwdrivers. These unidirectional screws shall be covered with transparent caps (not required for screw less design), ultrasonically welded with the meter body and the screw covers shall be embedded in the meter body in a groove. The meter shall withstand external magnetic influence as per latest amendments of CBIP Technical Report No.325 including 0.2T AC Magnet, 0.5T Permanent magnet.

7.0 TERMINAL BLOCK AND COVER

7.1 The terminals may be grouped in a terminal block having adequate insulating properties and mechanical strength. The terminal block shall be made from best quality non-hygroscopic, flame retardant material (capable of passing the flammability tests) with nickel plated brass inserts / alloy inserts for connecting terminals. It shall be rigidly fixed to the base of the meter so that it cannot be separated from the meter base without breaking either the meter base or the terminal block and this fixing arrangement shall be in parallel to the meter base in such a way that it cannot be viewed or approached from any part of the meter without breaking the meter.

7.2 The terminals in the terminal block shall be of adequate length in order to have proper grip of conductor. The screws shall have thread size not less than M4 and head having 6 mm diameter. The screws shall not have pointed ends at the end of threads. All terminals and connecting screws and washers shall be of tinned / nickel plated brass material. The terminal shall withstand glow wire test at 960 ± 15°C and the terminal shall withstand at least 135°C as per IS.

7.3 The internal diameter of terminal hole shall be minimum 9.5 mm and center to center distance shall be 13 mm. The holes in the insulating material shall be of sufficient size to accommodate the insulation of conductor also.

- 7.4 The terminal cover shall be transparent re-enforced Polycarbonate, Engineering Plastic with minimum thickness of 2.0 mm and the terminal cover shall be of extended type completely covering the terminal block and fixing holes. The space inside the terminal cover shall be sufficient to accommodate adequate length of external cables.
- 7.5 The terminals and all connecting screws will be of suitable material capable of withstanding a current of 150% of I_{max} for two hours, continuously and the meter shall be capable of providing phase to neutral protection up to 110 V for 1(one) hour.

8.0 MARKING OF THE METER

The marking on the meter shall be in accordance with relevant clauses of IS 14697. The basic marking on the meter nameplate shall be as follows. All other markings as per IS shall also be there.

- 8.1 Manufacturer's name & trade mark
- 8.2 Type Designation
- 8.3 No. of phases & wires
- 8.4 Serial number (Size not less than 5mm)
- 8.5 Month & year of manufacture
- 8.6 Reference Voltage
- 8.7 Rated Current
- 8.8 Operating Frequency
- 8.9 Principal unit(s) of measurement
- 8.10 Meter Constant (imp/Wh)
- 8.11 Class index of meter
- 8.12 "Property of WBSEDCL"
- 8.13 Purchase Order No. & Date
- 8.14 Guarantee (Guaranteed for a period of 5 ½ years from the date of delivery)
- 8.15 BIS marking
- 8.16 Place of manufacture
- 8.17 Barcode for meter serial no. in alpha numeric form, date of manufacture, current rating of the meter and PO reference, readable by single layer barcode reader.
- 8.18 The reference temperature if different from 27 °C.
- 8.19 The sign of Double Square for insulating encased meters

9.0 DISPLAY OF MEASURED VALUES

- 9.1 The meter shall have alphanumeric display with at least 7full digits with LCD backlit display, having minimum character height of 10 mm.
- 9.2 STN or TN or any better type of advanced LCD shall have to be used for visibility of display in poor light conditions. Back lit provided for clear visibility shall be uniform throughout all part of the LCD.
- 9.3 It shall be possible to easily identify the single or multiple displayed parameters through symbols or legends on the meter display itself or through display annunciation which shall be self explanatory and symmetric. Proper legends for the displayed parameters to be provided (Factory programmable).
- 9.4 Serial number of the meter shall be displayed on the display plate.
- 9.5 The meters shall have auto-display mode for pre-selected parameters. The push-button mode of display shall display all available parameters in meter and it shall have priority over auto-display mode.
- 9.6 The meter shall give clear message on display to indicate that the meter has experienced tamper along with the nature of tamper with date and time of first occurrence, last occurrence and last restoration. If the last tamper event is not restored, then meter will indicate first occurrence, last restoration and last occurrence.
- 9.7 Connection check, phase sequence and self diagnostics shall give clear message on display. The meter shall have a test output (blinking **LED**) accessible from the front and provision shall be there for monitoring the meter with suitable testing equipment.
- 9.8 The operation indicator must be visible from the front.
- 9.9 Test output device shall be provided in the form of one common LED for active and reactive energy with the provision of selecting the parameter being tested (separate LED may also be used with proper separation).

10.0 DISPLAY SEQUENCE

The meter shall display the required parameters in two different modes as follows. Display sequence for both auto and push button must be maintained, no interchange in sequence or display parameter will be accepted. All the display shall have proper legend for its easy identification.

10.1 Auto Display mode: The following parameters shall be displayed in auto cycle mode, in the following sequence:

- 10.1.1 LCD test
- 10.1.2 Meter Serial No.
- 10.1.3 Real Date (dd mm yy)
- 10.1.4 Real Time (hh mm ss)
- 10.1.5 Rising Apparent Demand with elapsed time (both Import & Export)
- 10.1.6 History1 Cumulative Active Import Energy (up to 24:00 hrs of last day of last month).
- 10.1.7 History1 Cumulative Active Export Energy (---do---)
- 10.1.8 Cumulative Active Import Energy
- 10.1.9 Cumulative Active Export Energy
- 10.1.10 Instantaneous Phase Voltages
- 10.1.11 Instantaneous Phase Currents
- 10.1.12 Instantaneous Power Factor
- 10.1.13 History1 Cumulative Reactive Import Energy(Lag) (up to 24:00 hrs of last day of last month)
- 10.1.14 History1 Cumulative Reactive Export Energy(Lag) ---do---
- 10.1.15 History1 Cumulative Reactive Import Energy (Lead) ---do---
- 10.1.16 History1 Cumulative Reactive Export Energy (Lead) ---do---
- 10.1.17 History1 Cumulative Apparent Import Energy ---do---
- 10.1.18 History1 Cumulative Apparent Export Energy ---do---
- 10.1.19 History 1 VA (Import)
- 10.1.20 History 1 VA (Export)
- 10.1.21 Cumulative Power OFF Hours
- 10.1.22 Power OFF Hours of present month
- 10.1.23 Cumulative Billing Count
- 10.1.24 Cumulative Tamper Count
- 10.1.25 Cumulative VA (Import) since manufacture
- 10.1.26 Cumulative VA (Export) since manufacture
- 10.1.27 Cover Open Information with date and time

10.2 Push Button mode: In addition to the auto display mode parameters, the following parameters shall be displayed on pressing the push button i.e. all display of auto mode and the display as per following:

- 10.2.1 Cumulative Reactive Energy Lag both import & Export, (up-to-date)
- 10.2.2 Cumulative Reactive Energy Lead both import & Export, (up-to-date)
- 10.2.3 Cumulative Apparent Energy both import & Export (up-to-date)
- 10.2.4 Present Month Active and Apparent Demand both import & Export with Date and Time
- 10.2.5 Cumulative Active Demand both import & Export since manufacture
- 10.2.6 Instantaneous Phase-wise Power Factor
- 10.2.7 Average Power Factor (Previous Month)
- 10.2.8 Average Load Factor (Previous Month)
- 10.2.9 Instantaneous Active Power
- 10.2.10 Instantaneous Apparent Power
- 10.2.11 Instantaneous Frequency
- 10.2.12 Present Tamper Status (PT/CT/Others)
- 10.2.13 First Occurrence of Tamper Event with Date & Time
- 10.2.14 Last Occurrence of Tamper Event with Date & Time
- 10.2.15 Last Restoration of Tamper Event with Date & time
- 10.2.16 Cumulative Power failure in hour: minute from the date of manufacturing
- 10.2.17 High resolution display for both Import & Export Wh, varh (Lag & Lead) and VAh (minimum 4+2)
- 10.2.18 High resolution display both for Import & Export phase-wise Wh (minimum 4+2)

- 10.2.19 Phase Sequence
- 10.2.20 Connection check(when all phases are forward or reverse)
- 10.2.21 Self- Diagnosis

10.3 Power Off Display : The following parameters shall be displayed during Power Off condition:

- 10.3.1 Meter Serial No.
- 10.3.2 Real Time (dd mm yy)
- 10.3.3 Date (hh mm ss)
- 10.3.4 History1 Cumulative Active Import Energy (up to 24:00 hrs of last day of last month)
- 10.3.5 History1 Cumulative Active Export Energy (---do----)
- 10.3.6 History 1 VA (Import)
- 10.3.7 History 1 VA (Export)
- 10.3.8 History 1 TOD wise Wh (Import)
- 10.3.9 History 1 TOD wise Wh (Export)
- 10.3.10 History 1 TOD wise VAh (Import)
- 10.3.11 History 1 TOD wise VAh (Export)
- 10.3.12 Cumulative Billing Count
- 10.3.13 Cumulative Tamper Count
- 10.3.14 Cumulative Active Import Energy
- 10.3.15 Cumulative Active Export Energy

10.4 Display parameters for auto and push button mode must be listed by the following two headers

- 10.4.1 "Auto Display Mode"
- 10.4.2 "Push Button Mode" (Parameters shall be pasted in front of the PP Box)

10.5 Display: Other requirements:

- 10.5.1 Each parameter shall be on meter display for 10 sec and the time gap between two auto display cycles shall be 30 sec.
- 10.5.2 The register shall be able to record and display starting from zero, for a minimum of 1500 hours, the energy corresponding to rated maximum current at reference voltage and unity power factor. The register shall not roll over in between this duration.
- 10.5.3 High resolution display can be given in separate mode and Energy in Wh shall be displayed in 4+2 digits, i.e, with 2 digits after decimal.
- 10.5.4 Up and down scrolling facility shall be there for Push button mode.
- 10.5.5 Push button mechanism shall be of high quality and shall provide trouble free service for a long span of time.

10.6 Besides other details the following parameters shall be available in BCS through downloaded data:

- 10.6.1 History 1 TOD wise Wh (Import)
- 10.6.2 History 1 TOD wise Wh (Export)
- 10.6.3 History 1 TOD wise VAh (Import)
- 10.6.4 History 1 TOD wise VAh (Export)
- 10.6.5 History 1 TOD wise MD in VA (Import)
- 10.6.6 History 1 TOD wise MD in VA (Export)
- 10.6.7 TOD Wise Cumulative Active Forwarded Energy both import & Export (up to date Zone 1, 2, 3)
- 10.6.8 TOD Wise Cumulative Apparent Energy bothimport & Export (up to date Zone 1,2,3)

11.0 MEASUREMENT OF HARMONICS

The meter shall be capable of measuring fundamental as well total energy i.e., fundamental plus harmonics energy. Total energy shall be made available on meter display and the same shall be used for billing purpose. Provision for measuring fundamental energy shall be kept for utilization in future. The total energy and fundamental energy shall be logged in the meter memory and can be downloaded through CMRI and made available in BCS.

12.0 INFLUENCE QUANTITIES

The meter shall be designed and protected such that all external effects and influences shall not change its performance & shall work satisfactorily within guaranteed accuracy limits, as specified in IS 14697: 1999 / CBIP technical Report 325, under the presence of the following influence quantities.

- 12.1 Electromagnetic field
- 12.2 External magnetic field
- 12.3 Radio frequency interference
- 12.4 Vibration
- 12.5 Voltage variation (70% - 120% of $V_{ref.}$)
- 12.6 Frequency variation ($\pm$) 5% of 50 Hz

13.0 RESETTING OF MAXIMUM DEMAND

- 13.1 The meter shall be capable of recording the Apparent MD with integration period of 15 minutes
- 13.2 MD reset shall be through each of the three means:
 - 13.2.1 Automatic resetting at preset date & time (at present it will be at 00.00 hrs of the first day of the month)
 - 13.2.2 Manually i.e., by push button
 - 13.2.3 Through authenticated command from CMRI or through Remote Communication
- 13.3 The means by which the reset has been done shall be made available to downloaded data. Facility to invoke any of the above through authenticated MRI command shall be provided at BCS.
- 13.4 MD reset button shall have proper sealing arrangement.
- 13.5 There shall be separate Push button for scrolling display (up and down) and MD reset. If only two Push buttons are used, minimum 20sec pressing is required for MD reset.

14.0 LOAD SURVEY

The meter shall be capable of recording load survey for the following parameters for a period of minimum 30 days - subject to availability of all parameters listed below with 15 minutes integration period.

- 14.1 Energy in Wh&VAh – phase-wise(both Import & Export)
- 14.2 Voltage – phase-wise
- 14.3 Current – phase-wise

15.0 NON VOLATILE MEMORY & REAL TIME CLOCK BATTERY

The Non Volatile Memory (NVM) shall not require any additional battery backup to retain the data in case of power failure at least for up to 10 years and the data storage shall be independent of battery backup unit. The life of the Real Time Clock (RTC) battery in circuit condition shall be minimum 10 years in case of power failure. It shall be possible to transfer this data to base computer software through CMRI/Laptop or RMR. The data so obtained shall be displayed in both graphical & numeric form in the BCS. The BCS with all details is to be provided by the supplier at no extra cost.

16.0 TIME OF DAY FACILITIES

The meter shall have facility to record Active & Apparent Energy (Import & Export) and MD (Import & Export) in at least 8 time zones. The time zones shall be user programmable through authenticated CMRI/Laptop/RMR command. Necessary software for the same is to be provided by the successful bidder. At present TOD timings will be programmed as follows:

- a. TOD 1: 06:00 Hrs to 17:00 Hrs.
- b. TOD 2: 17:00 Hrs to 23:00 Hrs.
- c. TOD 3: 23:00 Hrs to 06:00 Hrs.

17.0 METER READING DURING POWER OFF

It shall be possible to read the meter-display visually and with CMRI/Laptop in absence of input voltages with the help of internal battery backup. In case of external battery the arrangements shall be such that hands-free operation is possible. In case of external battery/power pack, 10 years' guarantee must be given for the same. Separate battery shall be used for this purpose(Not RTC or Processor battery).

18.0 SELF DIAGNOSTIC FEATURES

The meter shall have capability of performing self-diagnostic check to monitor the circuits for any malfunctioning to ensure integrity of data in memory location all the time. If possible, the details of malfunctioning shall be recorded in the meter memory. The bidder shall furnish the details of self-diagnostic capability feature, viz Memory status (NVM) and Battery status, RTC Status etc. and it shall be in display as well as in BCS.

19.0 MANUFACTURING ACTIVITIES

19.1 All the materials, electronics and power components, ICs used in the manufacture of the meter shall be of highest quality and reputed make to ensure higher reliability, longer life and sustained accuracy. The manufacturer shall use Application Specific Integrated Circuit (ASIC) or Micro controller for metering functions.

19.2 The electronic components shall be mounted on the printed circuit board using latest Surface Mounted Technology (SMT) except power components by deploying automatic SMT pick and place machine and re flow solder process. The electronic components used in the meter shall be of high quality **and there shall be no drift in the accuracy of the meter at least up to 10 years.**

19.3 Further, the Bidder shall own or have assured access (through hire, lease or sub-contract) of the mentioned facilities. The PCB material shall be of glass epoxy FR-4 grade conforming to relevant standards.

19.4 All insulating materials used in the construction of meters shall be non-hygroscopic, non-ageing and tested quality. All parts that likely to develop corrosion shall be effectively protected against corrosion by providing suitable protective coating. Quality shall be ensured at the following stages:

19.4.1 At PCB manufacturing stage, each board shall be subjected to bare board testing.

19.4.2 At insertion stage, all components shall undergo testing for conforming to design parameters and orientation. Complete assembled and soldered PCB shall undergo functional testing using test equipments (testing jig).

19.5 Prior to final testing and calibration, all meters shall be subjected to accelerated ageing test to eliminate infant mortality, i.e., meters are to be kept in ovens for 72 Hrs. at 55⁰C temperature & atmospheric humid condition. After 72 Hrs. meters shall work correctly. Facilities / arrangement for conducting ageing test shall be available with the manufacturer.

19.6 The calibration of meters shall be done in-house.

20.0 COMMUNICATION CAPABILITY

The meter shall have a galvanically isolated optical communication port as per IEC 1107/ANSI/PACT so that it can be easily connected to a hand-held common meter reading instrument (CMRI) for data transfer. The data downloading time shall be less than 5 minutes. The optical port shall be provided with proper sealing arrangement so that the optical cover shall not be opened without breaking the seal. The stored data in the meter shall be available through CMRI even when the display of the meter is not available.

A separate suitable serial port (RS-232/RJ11) capable of being hooked into a remote metering device such as modem, shall be provided inside the terminal cover to enable Automatic Meter Reading in such a way that the same cannot be accessed without opening the Terminal Cover and Seal.

Both the ports will have equal priority in terms of data accessing.

It shall not be possible to alter date in the meter bypassing commands from the CMRI or Laptop. Correction of RTC time, change of TOD timing, etc. shall be possible through CMRI/Laptop using authenticated command sets from BCS after scheduling for particular meter serial no. Billing parameters shall be factory programmable. No change shall be possible using CMRI only, i.e. the control has to be with the BCS.

21.0 BASE COMPUTER SYSTEM & SOFTWARE REQUIREMENTS

- 21.1 The Common Meter Reading Instrument (CMRI) shall be loaded with user-friendly MS-DOS 5.0 (or its higher version) compatible software for reading and/or downloading meter data.
- 21.2 Windows 7.0 (or its higher version) based Base Computer Software (BCS) shall be provided for receiving data from CMRI and downloading instructions from BCS to CMRI.
- 21.3 The data stored in the meter memory including defrauded energy shall be available on the BCS.
- 21.4 Only one BCS shall be provided for downloading data and authenticated command from CMRI/ Laptop.
- 21.5 The BCS shall have facility to convert meter reading data into user definable ASCII file format so that it can be integrated with the billing system or any other third party software. The user shall have the flexibility to select the parameters to be converted into ASCII file.
- 21.6 All the data available in the meter including energy, MD etc. with date and time stamp, new TOD time zones and historical data shall be available in BCS after downloading.
- 21.7 The bidder shall supply the necessary CMRI software during sample meter testing.
- 21.8 The bidder has to supply the meter reading protocol and API free of cost. The bidder shall indicate the relevant standard to which the protocol is compliant.
- 21.9 Transfer of data from the meter to CMRI & then to the BCS shall have to be easily executed.
- 21.10 Any change or up gradation of CMRI software or BCS in future, required for any reason, has to be done by the supplier at his own cost.
- 21.11 The same software shall be capable of preparing CMRI to read the meter information or to reconfigure the meter for change of TOD timings and / or time setting of the meter etc.
- 21.12 The exhaustive online help shall be available with the software so that user can use all the features of the software by just reading the help contents.
- 21.13 In BCS, data for Wh, VAh, MD & VA (total & TOD wise) for both Import & Export, average load factor, average power factor for at least twelve months must be available.

22.0 ANTI TAMPER FEATURES

The meter shall have the following anti-tamper features:

- 22.1 The meter shall work correctly irrespective of phase sequence of supply. There must be an indication of reversal of phase sequence in display as well as in downloaded data.
- 22.2 The meter shall work correctly even in absence of Neutral.

- 22.3 The meter shall be capable of recording energy correctly at Import as well as Export mode. If any phase current flows in forward direction, the meter shall register energy in phase-wise Import counter as well as in cumulative Import counter and if phase current flows in reverse direction, it shall register energy in phase-wise Export counter as well as in cumulative Export counter.
- 22.3 The meter shall record energy within maximum error of $\pm 4\%$ on injection of DC, pulsating DC (7-10 Hz), chopped AC in Neutral along with logging of ND tamper. In case tamper event is not logged, i.e, meter is immune to neutral disturbance, accuracy of the meter must not get affected. Maximum chopping for AC injection will be 25% to 30% at peak end.
- 22.4 The energy registration shall not be affected more than $\pm 4\%$ if high frequency (55Hz to 100Hz) or low frequency (45Hz to 30 Hz) AC signal w.r.to earth is applied to the meter neutral.
- 22.5 The meter shall be immune to Electro Static Discharge or sparks of 35 KV (approx) induced by using frequency-generating devices having very high output voltage.
- Tests in this respect will be conducted by using commonly available devices and during spark discharge test, spark will be applied directly at all vulnerable points of the meter for a period of 10 minutes (at an interval of 1 minute (approx) between two consecutive strokes) and meter shall maintain accuracy after the test under this condition. Accuracy will be checked during and after application of spark discharge Test. Meter shall record correctly within the specified limits of errors. Beyond 35 KV the meter shall record tamper if not immune.
- 22.6 The meter shall be capable of recording occurrence and restoration with date and time in respect to the following tamper events:
- 22.6.1 Power failure (Tamper count not to be increased)
 - 22.6.2 Invalid Voltage
 - 22.6.3 Missing Potential (phase wise)
 - 22.6.4 High Voltage
 - 22.6.5 Voltage Unbalance
 - 22.6.6 CT Open
 - 22.6.7 Invalid Phase Association
 - 22.6.8 Over Current
 - 22.6.9 Neutral Disturbances (If it is logged)
 - 22.6.10 Magnetic Disturbances
- 22.7 Threshold values of all above occurrence and restoration are attached elsewhere in this document. Snapshot values of Phase Voltage, Phase Current & Phase wise Power Factor, Active Energy value during occurrence & restoration to be provided in all the above mentioned tamper conditions in BCS with date and time. (In Event logging Snapshots shall be considered when the actual phenomenon occurred). The logging time for recording occurrence and restoration of all tamper events except Magnetic & Neutral Disturbance tamper, shall be 5 min. Magnetic tamper shall appear instantaneously, Neutral Disturbance within 3 min.
- 22.8 All authenticated commands shall be Base Computer Software (BCS) controlled. All transactions with meter shall be date and time logged, in the downloaded data (Last 12 months' transactions).
- 22.9 Properly designed meter tamper logic shall be provided and clearly explained in the bid. More than one tamper (CT related/ PT related/ others) shall not be logged at a time.
- 22.10 A minimum of 300 events (one event means either occurrence or restoration) of all types of tamper with date & time stamping shall be available in meter memory compartment wise.
- 22.11 The logging of events will be on FIFO basis. The events will be divided into three compartments like CT related (148 Events), PT related (88 Events) and others (64 Events).
- 22.12 Meter shall have a continuous and clear indication in its display if top cover is removed / opened and shall be logged in BCS as well as in display. It shall not be restored in any case.

23.0 GUARANTEE

23.1 The Meters and Pilfer Proof Meter Boxes shall be guaranteed for a period of **5½ years** from the date of supply against faulty design, poor materials, bad workmanship. The Meters and Pilfer Proof. Meter Boxes found defective within the above guarantee period shall have to be replaced by the supplier free of cost within one month on receipt of intimation. If the defective Meters and / or Pilfer Proof Meter Boxes are not replaced within the above specified period, WBSedCL will recover twice the cost of Meters and/or Pilfer Proof Meter Boxes from the supplier.

23.2 Life of battery used for the meter shall be guaranteed for **10 years**.

23.3 Name plate of the meter is to be marked with **“Guarantee of the Meter”: 5½ years from the date of supply.**

24.0 TYPE TESTING OF METER

The offered meters shall be type tested at any NABL accredited laboratory in accordance with IS 14697:1999 and CBIP Report 325 with their latest amendments. The type test report shall not be more than 3 (three) years old. A copy of the Type Test report shall have to be enclosed with the offer. If there is any modification in the design / parameters of the specifications or use of constituent materials in the offered meter submitted with the offer, from the meter which was submitted type tested, that may affect the characteristics as well as parameters of the meter, revised type test certificate as per the design, parameters and constituent material used in the offered meter, shall have to be submitted failing which the offer may be liable to be rejected. Type Test Certificate from any NABL accredited Lab shall only be considered. Type test certificate shall contain the following information clearly:

- i. Type of display or LCD
- ii. Class of accuracy
- iii. Meter constant
- iv. Type of meter

25.0 ACCEPTANCE TEST

The acceptance tests as stipulated in CBIP / IS (with latest amendments) shall be carried out by the supplier in presence of purchaser's representative. In case of failure of Meters as specified in Recommended Sampling Plan of IS-14697:1999, the entire lot will be treated as rejected. Also the following additional tests are to be carried out on one meter randomly selected from each lot offered for inspection / acceptance testing. In case of failure of any single meter the entire lot will be rejected.

- i. Magnetic induction of external origin (AC & DC)
- ii. Tamper & Fraud protection, as per relevant Clause of this specification.
- iii. Test of endurance up to 150% of I_{max} for two hours, followed by verification of limits of error
- iv. Verification of internal components
- v. Dry Heat Test under Test of Climatic Influences in IS 14697: 1999 of one meter from the offered lot is to be arranged by the supplier at any NABL accredited laboratory, at his cost

26.0 ROUTINE TEST

Each and every meter of the offered lot shall undergo the routine tests as well as functional tests as per IS: 14697:1999, CBIP Report 325 and after sealing the meters, the manufacturers will have to submit the routine test report of all the meters as well as a statement showing seal Sl. Nos. against each meter Sl. No. of offered lot in soft copy (MS WORD or EXCEL format), to the Chief Engineer(Procurement and Contract) and the Chief Engineer(DTD), along with offer letter for acceptance test.

27.0 TEST FACILITIES

- 27.1 The tests of equipment / instrument shall be carried out as per relevant Standards and test certificates shall be furnished for scrutiny. The Bidder shall indicate the details of the equipment available with him for carrying out the various tests as per relevant Standards. The bidder shall indicate the sources of all equipment / instruments.
- 27.2 The standard meters used for conducting tests shall be calibrated periodically at any NABL Accredited Test Laboratories and test certificates shall be available at Works for verification by purchasers' representative.
- 27.3 The manufacturer shall have at least the following testing facilities to ensure accurate calibration:-
- 27.3.1 AC high voltage test
 - 27.3.2 Insulation test
 - 27.3.3 Test of no load condition
 - 27.3.4 Test of Starting condition
 - 27.3.5 Test on Limits of error (Automatic Testing facility with ICT)
 - 27.3.6 Power loss in voltage and current circuit
 - 27.3.7 Test of Repeatability of error
 - 27.3.8 Test of meter constant
 - 27.3.9 Test of magnetic influence (As per CBIP 325 & Permanent Magnet)

28.0 INSPECTION

- 28.1 The purchaser may carry out the inspection at any stage of manufacture. The manufacturer shall grant free access to the purchaser's representative at a reasonable time when the work is in progress. Inspection and acceptance of any equipment under this specification by the purchaser shall not relieve the supplier of his obligation of furnishing the equipment in accordance with the specification and shall not prevent subsequent rejection if the equipment is found to be defective. All acceptance tests and inspection shall be made at the place of manufacturer unless otherwise especially agreed upon by the Bidder and purchaser at the time of purchase. The Bidder shall provide all reasonable facilities without charge to the inspector, to satisfy him that the equipment is being furnished in accordance with this specification.
- 28.2 The supplier shall keep the purchaser informed in advance, about the manufacturing programme for each lot so that arrangement can be made for inspection. The purchaser reserves the right to insist for witnessing the acceptance / routine testing of the bought out items. The supplier shall give 15 days for local supply / 30 days in case of foreign supply advance intimation to enable the purchaser to depute his representative for witnessing the acceptance and routine tests.
- 28.3 The purchaser reserves the right to get type test any meter, for meter casing etc. from any of the offered lots, reserve at any destination stores.

29.0 SUBMISSION OF SAMPLE METER

- 29.1 The bidder will submit his sample Meters in sealed casing / cartoon along with relevant Meter documents (**As per Annexure-IV**), on any working day, from 11.00 A.M. to 04.00 P.M. on weeks days & from 11.00 A.M. to 01.00 P.M. on Saturday within the specified period of submission latest by 01.00 P.M. on the last day of submission of bid to the Office of the Chief Engineer (DTD), Abhikshan, Sec-V, Salt Lake, Kolkata-91. The bidder will be given a receipt, jointly signed by the bidder and DTD officials, mentioning the samples and papers submitted by the bidder as per check list.
- 29.1.1 While submitting the samples and required documents as per Annexure-IV, the bidder shall submit three number of sealed meters as per the specifications stated herein before, 2 number of meters welded ultrasonically or by any other accepted technique and 1 number of meter , not welded, and another dummy meter case (for checking welding) and one P.P. Box.

- 29.1.2 They shall also submit one prototype of meter base and cover (with body screw caps) properly welded.
- 29.1.3 The date of testing of sample meters will be intimated to the bidders by CE (DTD) and during such test other bidders will also be allowed to witness the testing. Sample submission and Test procedure may be changed due to emergency requirement.
- 29.1.4 On the date of testing of sample meters of a particular bidder, he shall come prepared with the following: :
 - 29.1.4.1 BCS (as per specification)
 - 29.1.4.2 CMRI compatible with BCS and loaded with CMRI software
 - 29.1.4.3 laptop compatible with BCS
 - 29.1.4.4 Modem and accessories for testing the remote meter reading
 - 29.1.4.5 Any other accessories required for observing the performance and capabilities of the meters

30.0 QUALITY ASSURANCE PLAN

- 30.1 The design life of the meter shall be minimum 20 years and to prove the design life the firm shall have at least the following quality Assurance Plan.
- 30.2 The factory shall be completely dust proof.
- 30.3 The test rooms shall be temperature and humidity controlled as per relevant standards.
- 30.4 The test and calibrating equipments shall be automatic and all test equipment shall have their valid calibration certificates.
- 30.5 Meter will be tested (in case of lot test) in fully automatic test bench with ICT. No. human intervention will be allowed during test.
- 30.6 Power supplies used in test equipment shall be distortion free with sinusoidal wave forms and maintaining constant voltage, current and frequency as per the relevant standards.

31.0 CONNECTION DIAGRAM AND TERMINAL MARKING

Every meter shall be indelibly marked with a diagram of connection. For this poly-phase meter, this diagram shall also show the phase sequence for which the meter is intended. It is permissible to indicate the connection diagram by an identification figure in accordance with relevant standards. The marking of meter terminals shall appear on this diagram.

32.0 THE CHECKS TO BE CARRIED OUT DURING MANUFACTURING OF THE METERS

- 32.1 Meter frame dimensions tolerances shall be minimal.
- 32.2 The assembly of parts shall be done with the help of jigs and fixtures so that human errors are eliminated.
- 32.3 The meters shall be batch tested on automatic, computerized test bench and the results shall be printed directly without any human errors.

33.0 INFORMATION TO BE FURNISHED WITH THE BID

The Bidder shall invariably furnish the following information along with the bid, failing which the bid shall be liable for rejection. Information shall be separately given for individual type of material offered.

- 33.1 Statement giving list of important raw materials, names of sub-suppliers for the raw materials, list of standards according to which the raw materials are tested, list of tests normally carried out on raw materials.
- 33.2 Information and copies of test certificates in respect of bought out accessories.
- 33.3 List of manufacturing facilities available.
- 33.4 Level of automation achieved and lists of areas where manual processing exists.
- 33.5 List of areas in manufacturing process, where stage inspections are normally carried out of quality control and details of such tests and inspections.
- 33.6 List of testing equipment available with the bidder for final testing of equipment specified and test-plant limitations, if any, vis-à-vis type, special acceptance and routine tests specified in the relevant

standards and this specification. These limitations shall be very clearly brought out in schedule of deviations.

33.7 The list of components used in the meter.

33.8 A detailed list of bought-out items, which are used in the manufacturing of the meter indicating the name of firms from whom these items are procured. The details of quality assurance procedures followed in respect of the bought-out items.

33.9 The details of testing facilities available for conducting the routine and acceptance tests and other special tests on the meter.

33.10 The facility available if any for conducting type test.

33.11 Principle of operation of the meter, outlining the methods and stages of computation of various parameters starting from input voltage and current signals including the sampling rate, if applicable.

33.12 The relevant documents regarding the procurement of polycarbonate material.

34.0 LAB FACILITY: The laboratory of manufacturer must be well equipped for testing of the meters. They must have computerized standard power source and standard equipment calibrated not later than a year (or as per standard practice).

35.0 DOCUMENTATION

35.1 Twenty sets of operating manuals shall be supplied to the office of the CE (DTD) for distribution at sites.

35.2 One set of routine test certificates shall accompany each dispatch consignment.

35.3 The acceptance test certificates in case pre-dispatch inspection or a routine test certificate in cases where inspection is waived shall be approved by the purchaser.

36.0 REPLACEMENT OF DEFECTIVE METERS

The Meters declared defective by WBSEDCL within its guarantee period shall be replaced by the supplier with new ones free of cost within one month on receipt of intimation. Failure to do so within the prescribed time limit shall lead to imposition of penalty of twice the cost of meter. The same may lead to black listing even, as decided by WBSEDCL. In this connection the decision of WBSEDCL shall be final.

37.0 PACKING & FORWARDING

37.1 The equipment shall be packed in cartons / crates suitable for vertical / horizontal transport as the Case may be, and suitable to withstand handling during transport and outdoor storage during transit. The supplier shall be responsible for any damage to the equipment during transit, due to improper and inadequate packing. The easily damageable material shall be carefully packed and marked with the appropriate caution symbol. Wherever necessary, proper arrangement for lifting, such as lifting hooks etc., shall be provided. Supplier without any extra cost shall supply any material found short inside the packing cases immediately.

37.2 The packing shall be done as per the standard practice as mentioned in IS 15707: 2006. Each package shall clearly indicate the marking details (for e.g, manufacturer's name, Sl. Nos. of meters in the package, quantity of meter, and other details as per supply order). However, the supplier shall ensure the packing is such that, the material shall not get damaged during transit.

38.0 COMPONENT SPECIFICATIONS

The meters shall be designed and manufactured using SMT (Surface Mount Technology) components, except for power supply components, LED / LCD etc., which are PTH type. All the material and electronic power

components used in the manufacture of the meter shall be of highest quality and reputed makes so as to ensure higher reliability, longer life and sustained accuracy. The Components used for manufacture of meter shall be of high quality and the bidders shall confirm component specification as specified below in Annexure-III. Bidders shall compulsorily fill Annexure-I, Annexure-II & Annexure-III for technical qualification.

Sl. no.	Component Function / Feature	Requirement	Make / origin
1	Current Element	E-beam /spot welded C.T. shall be provided in the phase element and in the neutral with proper isolation.	Any make or origin conforming to IS-2705
2	Measurement / computing chips	The Measurement / computing chips used in the meter shall be with the Surface mount type along with the ASICs.	USA: Analog Devices, AMS, Cyrus Logic, Atmel, SAMES, Texas Instruments, Teridian; Japan: NEC, Freescale, Renesas; Holland: Phillips
3	Memory chips	The memory computing chips shall not be affected by the external parameters like sparking, high voltage spikes or electrostatic discharges.	USA: National Semi Conductor, Atmel, SAMES, Texas Instruments, Teridian, ST, Microchip; Japan: Hitachi, OKI, Freescale, Renesas; Holland / Korea: Phillips
4	Display modules	The display modules shall be well protected from the external UV radiations. The display shall be clearly visible over an angle of at least a cone of 70°. The construction of the modules shall be such that the displayed quantity shall not be disturbed with the life of display. The display shall be TN type industrial grade with extended temperature Range	Singapore: Bonafied Technologies; Korea: Advantek; Japan : Hitachi, SONY, Hijing, Truly Semiconductor; China: Tianma
5	Communication modules	Communication modules shall be compatible for the RS 232 ports	USA: National Semiconductors, HP, ST, Texas Instruments, Agilent, Avago; USA / Korea: Fairchild; Holland/ Korea: Philips; Japan: Ligitek, Hitachi, Germany: Siemens, Taiwan: Everlight,
6	Optical port	Optical port shall be used to transfer the meter data to meter reading instrument. The mechanical construction of the port shall be such to facilitate the data transfer easily.	USA: National Semiconductors, Texas Instruments, HP, Agilent, Avago, Germany/USA :Osram; Japan: Hitachi, , 21; Germany: Siemens; Holland / Korea: Philips; Taiwan: Everlight,
7	Power Supply	The power supply shall be with the capabilities as per the relevant standards. The power supply unit of the meter shall not be affected in case the maximum voltage of the system appears to the terminals due to faults or due to wrong connections.	As specified.
8	Electronic components	The active & passive components shall be of the surface mount type & are to be handled & soldered by the state of art assembly processes.	USA: National Semiconductors, Atmel, Phillips, Texas Instruments, ST, Onsemi; Japan: Hitachi, Oki, Toshiba. Freescale; Korea: Samsung.
9	Mechanical parts	The internal electrical components shall be of electrolytic copper & shall be protected from corrosion, rust etc. The other mechanical components shall be protected from rust, corrosion etc. by suitable plating / painting methods.	N.A.

10	Battery	Lithium-ion /Li with guaranteed life of 10 years	Renata, Panasonic, Varta, , Sanyo, National,Tadiran, Sony, Duracell, Tekcell, Mitsubishi, EVE, SAFT , XENO
11	RTC / Micro controller	The accuracy of RTC shall be as per relevant IEC / IS standards	USA: Dallas, Atmel, Motorola, NEC, Teridian, Renesas, Texas Instruments, ST, Micro chips, Epson; Holland / Korea: Philips; Japan: NEC, OKI, Hitachi, Mitsubishi, Freescale,
12	Pilfer Proof Meter Box	Technical specification given in separate sheet	Thermosetting Plastic

39.0 Annexure:

The Bidder shall submit the following schedules (as per Standard Format), which is part and parcel of the Specification.

Annexure-I: Guaranteed Technical Particulars of Meter

Annexure-I(b): Guaranteed Technical Particulars of Meter Box

Annexure-II: Pre-Qualification Criteria

Annexure-III: Component details

Annexure-IV: List of Documents to be submitted during sample submission

Annexure-V: List of Tests to be carried out on moulded sheet moulded compound meter boxes

Annexure-VI: Testing

TECHNICAL SPECIFICATION FOR PILFER PROOF METER BOX (SINGLE DOOR TYPE & WITHOUT CUT OUT) SUITABLE FOR 3 PHASE STATIC ENERGY METER

1.0 SCOPE :

- 1.1. This scope covers the manufacture and supply of Pilfer Proof Moulded Meter Box suitable to house 3-Phase Static Energy Meters. The Meter Box shall be suitable for wall mounted type.
- 1.2. Technical requirement and standard:- The meter box shall be made out of hot pressed moulded, unbreakable, high grade, fire retardant thermosetting plastic e.g. glass fibre reinforced polyester SMC (Sheet Moulding Compound) as per S1 grade of IS) 13410, or Glass Reinforced Polyester Dough Moulding Compounds as per Grade D1 of IS 13411, with flame retardant properties having good di-electric & mechanical strength. The Top cover of the Meter Box shall be as per enclosed drawing with provision of separate window arrangement as shown. The material must be U.V. stabilized to ensure that the Meter Box shall not change in colour, shape, size, dimension when subjected to 200 hrs. of U.V. Ageing Test. The Meter Box shall have top tapered surface / round corners to prevent stay of rain water at the top of the Meter Box.
- 1.3. The Meter Box shall be capable of withstanding the mechanical, electrical and thermal stress as well as the effects of humidity which are likely to be encountered in service. At the same time the same shall ensure the desired degree of safety. The plastic material used shall be adequately stabilized against detrimental effect of light and weather. The surface appearance of the moulded parts must be smooth, non-porous and homogeneous, free of ripples, defects and marks. No fillers or fibers shall be visible at any place. The Box shall comply in all respect with the requirement of latest Amendments of I.S. 13410-1992 Specification for "General, requirements for enclosures for accessories for fixing electric installations." Applicable degree of protection shall be I.P. 42 or better. The SMC material which will be used by bidder for this moulded Meter Box conforms to Relevant IS/13410-1992 with latest amendment.
- 1.4. The Enclosures shall generally comply with the provision of IS 14772 or IEC 695. The enclosures shall be suitable for outdoor application. The enclosure shall be with good workmanship. Soft neoprene/nitrile rubber gaskets shall be provided all round wherever required for protection against entry of dust and water. The gasket shall confirm to Type-III as per IS-11149. The enclosure shall comply with IP-54 degree of protection. The Enclosures shall be off- white/admiral grey/ivory or as specified by the owner.

2.0 GENERAL CONSTRUCTIONAL REQUIREMENT :

- 2.1. **Dimension:** Length - 400 mm, Breadth - 300 mm, Height - 200 mm. Thickness of the enclosure shall not be less than 2 mm on all sides including door. The enclosure shall have 4 mm thickness of the tongue and groove area. There shall be a minimum 30mm clearance on all sides from the meter surface (projected) except the bottom side which shall be minimum 75 mm from the lower edge of terminal block. Meter mountings inside the meter box will be such that the meter base support inside box shall be preferably raised by about 10 +/- 2 mm for each items for each of working. Fixing arrangement of meters and other equipment to the base of the meter box shall be as per provision of the drawing and as per the specification.
- 2.2. **Viewing Window:** A viewing window (175 x 85 mm with tolerance of 5 mm or as per requirement of the owner) made up of scratch and break resistant, UV resistant, transparent Polycarbonate / toughened glass shall be provided on the door for reading the meter without inconvenience. The minimum thickness of the viewing window shall be 2.0 mm. The window shall be securely fixed with meter enclosure from inside Suitable neoprene gasket shall be provided so that there shall not be any ingress of moisture through this window into the meter box.
- 2.3. **Hinges:** A minimum of 2 nos. brass/stainless steel hinges on each door shall be provided inside the enclosure. The hinges of the door shall be concealed and they shall be fixed to the flanges provided on the body and cover of the enclosure in such a manner that the door opens by a minimum of 120 degrees.
- 2.4. **Locking Arrangement:** The cover shall be fitted with base and shall be of concealed hinges. It shall have some knobs provided with covers. The covers are to be fixed on the base of Meter Box in such a way that any access from outside is not possible. There shall be provision of padlock simultaneously with holes for sealing arrangement covering the top of the Meter Box. The door shall be provided with SS latch or U clamp similar

cable entry holes shall be provided in the bottom of the Meter Box as per the drawing enclosed and the intermediate partition plates.

- 2.5. Sealing Arrangement:** The meter box shall have provision for minimum 2 nos. seals to make it fully tamper proof.
- 2.6. Arrangement for Inlet & Outlet Cable Entry:** Two (2) nos. circular holes having 40 mm dia shall be provided at the bottom of the Meter Box for inlet and outlet of armoured aluminium cables. High grade double compression MS cable glands fixed on both sides by check nuts are to be provided for securely fixing the cable at the bottom of meter box. A suitable arrangement like clamping nut may be provided with the gland so that opening dia can be reduced to the size of Cable.
- 2.7. Base and cover details:** The cover shall be made overlapping type having collars on all 4(four) sides. The cover of the Meter Box shall be provided with Gasket of sufficient size to completely fit in the grooves of the base. The gasket shall be made of high quality neoprene rubber. The base of the Meter Box must have a groove to hold the Gasket and the overlap of the top cover with base must be minimum 6 mm.
- 2.8. Fixing arrangement of Meter Box:** For fixing the Meter Box to wall or wooden board, 4 no. holes (two top side holes to be keyholes) of minimum 4 mm dia shall be provided at the four corners of meter cup-board. The meter is to be installed in the Meter Box and the Box in assembled condition shall have provision to fix it on wall. The 4 no. self threaded screws of min. size of 4 mm dia and 25 mm long shall be provided with each Meter Box.
- 2.9. Marking / Embossing:** The following information shall be clearly & indelibly embossed on the cover and base of the Meter Box or printed on metallic plate and duly riveted on the box cover. The top & bottom corner of Meter Box Sl.No. shall be same for the particular Meter Box.
- 2.9.1.** Property of WBSEDCL
 - 2.9.2.** Name / Brand name of Manufacturer
 - 2.9.3.** Meter Box Sl. No. (Embossed on both the base & covers of Meter Box)
 - 2.9.4.** Sign of danger
- 2.10. Drawing:** Detailed dimensional drawing showing clearly the dimensions & material for Meter Box and its constructional features have been furnished with the tender specification which is binding on the part of the manufacturer.

3.0 SUBMISSION OF SAMPLE:

- 3.1.** Bidder shall submit a sample Meter Box as per our specification (or, as available with Bidder, however in the event of order it will be as per Specification before first inspection) along with the sample meter to the office of the Chief Engineer, (DTD), AbhikshanBhavan, Sector-V, Salt Lake, Kolkata-91 before the last day of submission of bid.
- 3.2.** Submission of sample meter box as per size available with the bidder but conforming to our specification towards its quality is acceptable. Type testing including material identification (IR Spectrometry test) of one meter box manufactured as per specification is to be conducted at any NABL accredited laboratory/CIPET by the supplier at their own cost after placement of
- 3.3.** order. For type testing the meter box will be selected from the first offered lot of meter with meter box. If the type test results are not found satisfactory, the offered lot of meter along with meter box will be rejected.

4.0 TESTING:

- 4.1. Type Test:** The Meter Box offered shall be type tested as per relevant I.S. standards and the Technical Specification stated below : The bidder must furnish type test report including material verification of the offered /sample meter box from any NABL/Govt. approved laboratory as available with them along with technical bid without which the offer will not be considered. Type test report shall not be more than 5 (five) years old.
- 4.2. Acceptance Test:** The acceptance test as indicated in the enclosed table (Annexure-V) shall be carried out at the time of inspection of the offered material.

4.3. Routine Test: The routine tests as indicated in the enclosed table (Annexure-V) shall be carried out and routine test certificates / reports shall be submitted to the purchaser's inspection office at the time of inspection of the offered material enclosed table (Annexure-V) shall be carried out at the time of inspection of the offered material.

5.0 GUARANTEE:

5.1. The Pilfer Proof Meter Box shall be guaranteed against any manufacturing defects arising out of faulty design or bad workmanship or component failure for a period of 5 ½ years from the date of supply. The meter box found defective within the above guarantee period shall be replaced by the Supplier free of cost within one month of the receipt of intimation of failure/defect. Defective meter box are to be replaced by new one with new sl. nos. as allotted by C.E (DTD).

5.2. Replacement of defective Meter Box : The Meter Box declared defective by the WBSEDCL shall be replaced by the supplier up to the full satisfaction of the WBSEDCL at the cost of supplier as per terms of GCC within 60 (sixty) days from the date of intimation by the purchaser. Failure to do so within the time limit prescribed shall lead to imposition of penalty of twice the cost of meter box. The same may lead to black listing even, as decided by WBSEDCL. In this connection the decision of WBSEDCL shall be final.

6.0 INSPECTION: The inspection will be carried out as per inspection & testing clause of General Conditions of Contract (GCC).

7.0 GUARANTEED TECHNICAL PARTICULARS: The bidder shall furnish all the necessary information as per Annexure-VI – Guaranteed Technical Particulars. If the bidder desire to furnish any other information in addition to the details as asked for, the same may be furnished.

Annexure-I

GUARANTEED TECHNICAL PARTICULARS FOR 11KV STATICTRIVECTOR BOUNDARY METERS

Sl. No.	Item Description	Required Specification
1	Name of manufacturer	Bidder to specify
2	Type (Model No.), name, number	Bidder to specify
3	Standards Applicable	IS 14697: 1999 read with latest amendments, IS 15707, IS 13410:1992 along with latest amendments, IS 12346:1988, CBIP Report No. 325 & its latest amendments
4	Ratings	
(i)	Accuracy class	0.5S
(ii)	Rated Voltage	Ph-N : 110 / $\sqrt{3}$ V (63.5 V)
(iii)	Basic Current (I_b)	-/ 1 A
(iv)	Rated Frequency	50 Hz $\pm$ 5%
(v)	Power Factor Range	0 Lag-unity-0 Lead
(vi)	Meter Constant (imp/Wh)	Bidder to specify
5	Minimum Starting Current	0.1% of I_b
6	Power Loss at Rated Frequency & Reference Temperature in :	
(i)	Current Circuit	Less than 1.0 VA
(ii)	Voltage Circuit	Less than 1.5W & 8VA
7	Allowable Voltage Variation	70% to 120% of Rated Voltage
8	Parameters displayed	To be mentioned
9	Dynamic range	As per IS 14697
10	MD reset Provisions	Provisions to reset MD: a) Using CMRI through authenticated Command b) Using Remote Communication Command b) Automatic Resetting at preset date & time c) Manual Resetting through push button
11	Display :	

(i)	Type of Register	LCD suitable for operation up to 60°C
(ii)	No. of Digits of Display	8 Full Digits (minimum)
(iii)	Height of character	10 X 5 mm (minimum)
(iv)	Auto display mode & scroll mode	Both required
(v)	Type of Push Button for scroll mode	Spring loaded push button to be provided on top cover of meter
12	Non Volatile Memory	To be provided as per IS specification
13	Provision for taking reading during power off condition	During power off condition also, meter shall display reading with backup power through the push button provided on the meter. All data downloading facility shall have to be provided in power off condition
14	Principle of operation	As per applicable standards
15	MD Integration period	15 min
16	Weight of meter	Bidder to specify
17	Dimension	Bidder to specify
18	Guarantee	5 & ½ years from the date of supply
19	Outline drawings & Leaflets	Bidder to provide
20		
(i)	Remote meter readout facility	Must have provision
(ii)	Communication Protocol used	DLMS Cat B
(iii)	Sealing provision for Meter & Optical Port	At least two fixing screws for fixing meter cover with the meter base shall be provided. Each Screw shall have two holes; one for fitting/fixing of manufacturer's seal and the other for utility seal. The arrangement shall be in such a manner that any access to the working part of the meter body will not be possible without breaking/ tampering / removing the said seals. Suitable arrangement shall be made for fitting/fixing of utility seal at two sides of meter terminal cover in such a manner that any access to the terminal cannot be possible without removing the seals. There shall also be provision for sealing the optical port. The meter cover shall be permanently fixed to the base by ultrasonic welding or by any other technic which is either equally or more efficacious so that cover cannot be opened without breaking, i.e. the meter shall be break to open type. Meter shall have an indication in its display if top cover is removed.
(iv)	Baud rate of data transmission	9600 bps
(v)	Required software to be resident in HHU	Bidder to provide
(vi)	Ultrasonic welding (or any other technology, equally or more efficacious) of body	Must be provided
(vii)	Manufacturer's Seal	Must be provided
21	Base Computer Software	Compatible with Windows 7 or higher
22	Type Test Certificates	Bidders to provide
23	Time of Day Zones (Selectable)	3 Zones to be provided with provision of 8 Zones
24	Whether meter measures both fundamental & harmonic energy	As per specification
25	Real Time Clock Accuracy	Maximum drift ± 5 min per annum
26	Battery for Real Time Clock	It shall be Lithium-ion / Lithium battery having at least 10 years of life . In case of Li battery the rating shall be more than 500mAh.
27	Anti Tamper Features	Must be as per Tamper Logic provided by WBS EDC
28	Effect of Accuracy under tamper condition	
29	Drift in Accuracy of measurement with time	No Drift with time
30	Name plate details	As per relevant IS and specified Technical Specification
31	Type of calibration	Meter shall be software calibrated at factory & there

		shall not have any provision for mechanical form of calibration
32	Type of mounting	Projection Type
33	Testing facility	
(i)	Fully automatic computerized meter test bench with print out facility shall be available	Must be available
(ii)	Make and Sl. No. of Test bench with calibration validity	Bidders to specify
(iii)	Accuracy of ESS duly calibrated	Bidders to specify
(iv)	Following in house testing facility shall be available:	
	(i) AC high voltage test	Must be available
	(ii) Insulation test	Must be available
	(iii) Test of no load condition	Must be available
	(iv) Test of Starting condition	Must be available
	(v) Test on Limits of error	Must be available
	(vi) Power loss in voltage and current circuit	Must be available
	(vii) Test of Repeatability of error	Must be available
	(viii) Test of meter constant	Must be available
	(ix) Power loss in voltage & current circuit	Must be available
	(x) Test of Magnetic influence	Must be available
34	Data retention by NVM without battery backup and under power off condition	10 years
35	Type of material used	
(i)	Base	High impact strength, non-hygroscopic, fire retardant, fire resistant, UV stabilized poly carbonate (LEXAN 503R or equivalent). Meter base with LEXAN 500R/143R may be accepted subject to verification
(ii)	Cover	High impact strength, non-hygroscopic, fire retardant, fire resistant, UV stabilized transparent poly carbonate (LEXAN 943A or equivalent)
(iii)	Terminal Block	Material High impact strength not hygroscopic, fire retardant, fire resistant, UV stabilized poly carbonate (LEXAN 503R or equivalent) Barrier of adequate size shall be provided between phase and neutral. Terminal block with LEXAN 500R/143R may be accepted subject to verification
(iv)	Terminal Cover	High Impact strength, non-hygroscopic, fire retardant, fire resistant, glass reinforced poly carbonate (Transparent) & non detachable with hinging arrangement (LEXAN 943A or equivalent). Terminal cover with LEXAN 143 may be accepted subject to verification
36	Screw	
(i)	Material	Tin/Nickel-plated Brass
(ii)	Size	As per relevant IS Standards
37	Internal Diameter of Terminal Hole	9.5 mm (minimum)
38	Center to Center clearances between adjacent terminals	13 mm
39	Security Profiles	
(i)	Basic Security	To be provided
(ii)	Advance Security	To be provided

Annexure-I(b)

GUARANTEED TECHNICAL PARTICULARS OF THREE PHASE METER BOX:

Sl. No.	Item Description		Manufacturer's Particular
1	Name & address of manufacturer		
2	Material		Thermosetting Plastic
3	Grade of Material		SMC / DMC Ref. Standard IS : 13410 – 1992
4	Properties of material for meter box		
a)	Heat Deflection Temperature (Ref. Std. IS: 13411)		150 °C (minimum)
b)	Exposure to flame (Ref. Std. IS: 4249)		Self-extinguishing
c)	Melting Point (Ref. Std. IS: 13360)		Shall not melt above 180 °C
d)	Tensile Strength (MPa)		50 Mpa (minimum)
e)	Flexural Strength (MPa)		90 Mpa (minimum)
f)	Modulus of Elasticity (MPa)		2000 Mpa (minimum)
g)	Izod impact strength notched 230 °C		8 KJ/Sq M (minimum)
5	Constructional features of the box		
(a)	Clear inside dimensions of Meter Box		Refer Drawing
	i.	Height	400 mm
	ii.	Width	300mm
	iii.	Depth	200 mm
	iv.	Rust & Vermin proofing	Neoprene Rubber Gasket NRG
(b)	a)	Material of transparent cover	Toughened glass / transparent polycarbonate with rubber gasket
	b)	Size of opening/viewing window (Min.)	175mm x 85 mm with tolerance of 5 mm
	c)	Minimum thickness of cover	2 mm
	d)	Fixing method	Fixed from inside with rubber gasket
(c)	Sealing Arrangement		Holes for wire seal (2 Nos.)
(d)	Wire entry		MS / Aluminium glands fixed on both side / bottom side by check nuts to be provided.
(e)	Colour of Meter Box		Grey / Off White
(f)	Meter mounting arrangement		Suitable for mounting of any make of meter (broadly as per the drawing)
(g)	Box mounting arrangement		4 number of holes
(h)	Hinges		2 number of concealed hinges or better arrangement
(i)	Incoming & outgoing cable holes		2 number of 40mm diameter holes and bottom entry. High grade double compression MS / Aluminium Cable glands fixed on both sides by check nuts to be provided.

Annexure-II
Pre-Qualification Conditions for Three Phase Static Meters

Sl. No.	Requisite Qualification	Remarks
1	Bidder shall have valid BIS certification for the offered meter	Yes / No
2	Bidder shall have ISO 9001 certification	Yes / No
3	Bidder shall be manufacturer of static meters having supplied Static 1-phase or 3-phase meters with memory and LCD display to Electricity Utilities in last 2 years	Yes / No
4	Bidder shall have Type Test certificate from any NABL accredited laboratory for the type of offered meter not older than three years	Yes / No
5	Bidder shall have dust free, static protected environment for manufacture, assembly and testing of meters	Yes / No
6	Bidder shall have automatic computerized test bench for lot testing of meters	Yes / No
7	Bidder shall have facilities of Oven for ageing test	Yes / No
8	Bidder shall have certificate from any NABL accredited laboratory for immunity of the type of meter offered, against magnetic influence of 0.2 T AC & 0.5 T DC	Yes / No

Annexure-III

Sl. No.	Component Function / Feature	As per Requirement	Make / origin
1	Current Element		
2	Measurement / Computing chips		
3	Memory chips		
4	Display modules		
5	Communication modules		
6	Optical port		
7	Power Supply		
8	Electronic components		
9	Mechanical parts		
10	Battery		
11	RTC / Micro controller		

Annexure-IV

Sl.No.	LIST OF DOCUMENTS TO BE SUBMITTED DURING SAMPLE SUBMISSION
1	Attested copy of type test reports from NABL accredited laboratory
2	Attested copy of BIS certificates of the same type of meter submitted as sample
3	Attested certificates as regards material used for meter case, cover & terminal block
4	Annexure – II as per tender document
5	Annexure – III as per tender document
6	Operating manual of the meter submitted

Annexure-V

LIST OF TESTS TO BE CARRIED OUT ON MOULDED SHEET MOULDED COMPOUND METER BOXES

Sl no.	Name of Indian Standard equivalent international	Clause Ref.	Test requirement	Test particulars		
				Type	Routine	Acceptance
1.	IS : 14772		Marking	T		A
2.	As per Co.'s Drg.	Clause 9	Dimensions	T	R	A
3.	IS : 14772	Clause 9	Protection against electric shock	T	R	A
4.	IS : 14772	Clause 02	Construction	T	R	A
5.	IS : 14772	Clause 02	Resistance to ageing, to humid condition, to ingress of solid object and to harmful ingress of water	T		
6.	IS : 14772	Clause 02	Mechanical strength	T		
7.	IS : 14772		Resistance to heat	T		
8.	IS : 14772		Resistance to tracking	T		
9.	IS: 8620/1996		Test for resistance to heat & fire. Glow wire test at 650°C as per cl.4 to 10 of IEC 695-2-1.	T		
10.	IS:13411		Heat deflection temp. (Min.1500C.)	T		
11.	IS :4249		Self Extinguishing property of spirit burner test.	T		
12.	IS / ASTM		Melting point- 180°C (Min)	T		
13.	IS :8623/1993	(Part-1) 18.2.2.2	Verification of dielectric properties, insulation test with 500V DC Megger.	T		
14.	CIPET/IR Spectrometry		Material identification	T		
15.	IS :13411/1992		Test for water absorption (Max. 0.35%)	T		

Note: Applicable degree of protection shall be IP 42 or better.

Legend: T=Type Test, R=Routine Test, A=Acceptance Test

Annexure-VI

TESTING:

1. Sample selected from first lot shall be tested at Govt. approved independent test house for compliance of performance parameters as given in GTP including material identification to be carried out by CIPET/IR Spectrometry.
2. The test report shall be submitted to WBSEDCL before completion of order.
3. Inspection of each lot, sampling plans for test: 1 nos. selected randomly from lot for testing at works.

Sl. no.	Test Requirement for moulded meter box	Reference Standards
(a)	Marking	IS:14772
(b)	Dimensions & construction	IS:14772
(c)	Heat Deflection Temperature (Min. 150°C)	IS:13411
(d)	Spirit Burner Test (Self Extinguishing)	IS:4249
(e)	Melting Point (Does Not Melt up to 400 deg. C.)	IS:13360